2008 GENERAL INFORMATION

Fuel System - General Information - Edge & MKX

SPECIFICATIONS

MATERIAL

Item	Specification	Fill Capacity
Motorcraft SAE 5W-20 Premium Synthetic Blend Motor Oil XO-5W20-QSP (US); Motorcraft SAE 5W-20 Super Premium Motor Oil CXO-5W20-LSP12 (Canada); or equivalent	WSS-M2C930-A	-

GENERAL SPECIFICATIONS

Item	Specification
Fuel tank capacity (all wheel drive [AWD] vehicles)	76L (20 gal)
Fuel tank capacity (front wheel drive [FWD] vehicles)	72L (19 gal)
Fuel Pressure	
Engine running fuel pressure	450 kPa (65 psi)

DESCRIPTION AND OPERATION

FUEL SYSTEM

WARNING: **Do not smoke, carry lighted tobacco or have an open flame of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.**

WARNING: **Do not carry personal electronic devices such as cell phones, pagers or audio equipment of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.**

WARNING: **Before working on or disconnecting any of the fuel tubes or fuel system components, relieve the fuel system pressure to prevent accidental spraying of fuel. Fuel in the fuel system remains under high pressure, even when the engine is not running. Failure to follow this instruction may result in serious personal injury.**

WARNING: When handling fuel, always observe fuel handling precautions and be prepared in the event of fuel spillage. Spilled fuel may be ignited by hot vehicle components or other ignition sources. Failure to follow these instructions may result in serious personal injury.

The fuel system consists of:

- a mechanical returnless fuel system.
- a fuel rail.
- a multi-port fuel injection (MFI) system.
- an all wheel drive (AWD) saddle-type or front wheel drive (FWD) L-shaped fuel tank.
- fuel, vapor and brake tubes in an integrated bundle assembly.
- a fuel tank filler pipe assembly, which cannot be modified in any way.
- a 1/4 turn-type fuel tank filler cap.
- an inertia fuel shutoff (IFS) switch.
- a separate fuel level sensor assembly on AWD saddle-type fuel tank.
- a fuel pump (FP) module containing:
    - the electric FP module, which provides pressurized fuel to the fuel rail.
    - a serviceable fuel level sender.
    - a check valve, which maintains system pressure after the FP module is shut off.
    - a pressure relief valve for overpressure protection in the event of restricted fuel flow.
    - an in-tank fuel filter providing filtration to protect the fuel injectors from foreign material.

The FP module is controlled by the PCM. Electrical power to the FP is provided through the IFS switch located under the left rear quarter trim panel.

The engines are equipped with sequential MFI injection.

The fuel pressure is regulated to a constant pressure by a fuel pressure regulator.

DIAGNOSTIC TESTS

FUEL SYSTEM

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
 ST2311-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	-

Principles of Operation

NOTE: **The following procedure diagnoses a slow to fill concern only. For all other concerns, refer to the INTRODUCTION - GASOLINE ENGINES article.**

The fuel tank filler pipe assembly is used to refuel the vehicle. The fuel tank filler pipe flapper valve prevents spitback of fuel during and after refueling. The fuel tank stores the fuel. The fuel tank contains a fuel pump (FP) module. The FP module consists of a fuel level sender and a FP. The fuel level sender sends a signal to the fuel gauge informing the driver of how much fuel is in the fuel tank. The FP provides fuel to the fuel tubes which supply the fuel rail.

During refueling, the fuel tank vents to the atmosphere through the vent and filler pipes, on vehicles without on-board refueling vapor recovery (ORVR) systems. In vehicles equipped with ORVR, the fuel tank and filler pipe are designed so that when the vehicle is being refueled, fuel vapors in the fuel tank travel to the evaporative emission (EVAP) canister, which absorbs the fuel vapors and vents the pressure from the fuel tank during refueling.

Inspection and Verification

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1. Verify the customer's concern by refueling the vehicle and observe the fuel fill rate.
2. Inspect to determine if any of the following mechanical concerns apply.

VISUAL INSPECTION CHART

Mechanical
• Bent, kinked or damaged fuel tank filler pipe • Bent, kinked or damaged fuel tank filler pipe vent tube (if equipped) • Incorrect routing of the fuel tank filler pipe

- Incorrect routing of the fuel tank filler pipe vent tube (if equipped)
- Incorrect position of fuel tank filler pipe clamps
- Incorrect position of fuel tank filler pipe vent tube clamps (if equipped)
- Fuel tank mounted vapor tubes bent or damaged
- Evaporative emission (EVAP) system tubes or hoses bent or damaged
- Accident damage to the fuel tank
- Accident damage to the vehicle effecting the fuel tank filler pipe-to-body connection
- Unauthorized modifications and/or alterations to the vehicle
- EVAP system fresh air tube plugged (dirt, spider webbing)

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

NOTE: **Make sure to use the latest scan tool software release.**

4. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: **The vehicle communication module (VCM) LED prove out confirms power and ground from the DLC are provided to the VCM.**

5. If the scan tool does not communicate with the VCM:
   - check the VCM connection to the vehicle.
   - check the scan tool connection to the VCM.
   - refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
6. If the scan tool does not communicate with the vehicle:
   - verify the ignition key is in the ON position.
   - verify the scan tool operation with a known good vehicle.
   - refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
7. Carry out the network test.
   - If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
   - If the network test passes, retrieve and record continuous memory DTCs.
8. Clear the continuous DTCs and carry out the self-test diagnostics for the EVAP system.
9. If the DTCs retrieved are related to the concern, go to **EVAPORATIVE EMISSION SYSTEM DTC**

CHART. For PCM related DTCs, refer to the **INTRODUCTION - GASOLINE ENGINES** article. For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.

10. If no DTCs related to the concern are retrieved, go to **SYMPTOM CHART**.

EVAPORATIVE EMISSION SYSTEM DTC CHART

DTC	Description	Action
P0446	Evaporative Emission System Vent Control Circuit	Go to **PINPOINT TEST A**.
P0451	Evaporative Emission System Pressure Sensor/Switch Range/Performance	Go to **PINPOINT TEST A**.
P0452	Evaporative Emission System Pressure Sensor/Switch Low	Go to **PINPOINT TEST A**.
P0453	Evaporative Emission System Pressure Sensor/Switch High	Go to **PINPOINT TEST A**.
P0454	Evaporative Emission System Pressure Sensor/Switch Intermittent	Go to **PINPOINT TEST A**.
P1443	Evaporative Emission System Control Valve (Low/No Flow)	Go to **PINPOINT TEST A**.
P1450	Unable to Bleed up Fuel Tank Vacuum	Go to **PINPOINT TEST A**.
P1451	Evaporative Emission System Vent Control Circuit	Go to **PINPOINT TEST A**.
P260F	Emission System Monitoring Processor Performance	Go to **PINPOINT TEST A**.

Symptom Chart

SYMPTOM CHART

Condition	Possible Sources	Action
• Slow to fill	• Fuel tank filler pipe • Fuel tank filler pipe vent tube, if equipped • Evaporative emission (EVAP) system • Fuel tank filler pipe flapper valve (part of the fuel tank filler pipe assembly) • Fuel level vent valve, if equipped (part of the fuel tank)	• Go to **PINPOINT TEST A**.
• All other fuel system concerns	• Fuel system components	• REFER to the **INTRODUCTION - GASOLINE ENGINES** article.

Pinpoint Test A

Pinpoint Test A: Slow to Fill

Normal Operation

Under normal operation, fuel should flow at a steady rate through the fuel tank filler pipe into the fuel tank. As fuel enters the fuel tank, air is vented through the filler pipe or the on-board refueling vapor recovery (ORVR) system.

This pinpoint test is intended to diagnose the following:

- Fuel tank filler pipe vent tube, if equipped
- Fuel tank filler pipe
- Evaporative emission (EVAP) system
- Fuel tank filler pipe flapper valve (part of the fuel tank filler pipe assembly)
- Fuel level vent valve, if equipped (part of the fuel tank)

PINPOINT TEST A: SLOW TO FILL

A1 CARRY OUT INSPECTION AND VERIFICATION

- Carry out inspection and verification.
- **Was the cause of the concern found?**

  YES : REPAIR or INSTALL new components to correct the concern.

  NO : Go to A2.

A2 CHECK THE SYSTEM FOR ANY EVAP DTCs

- Connect the scan tool.
- Check the system for any of the following EVAP DTCs: P0446, P0451, P0452, P0453, P0454, P1443, P1450, P1451 and P260F.
- **Are any of these DTCs present?**

  YES : REFER to **INTRODUCTION - GASOLINE ENGINES** article to diagnosis the EVAP system.

  NO : Go to A3.

A3 MONITOR THE FUEL TANK PRESSURE (FTP) WHILE FILLING THE FUEL TANK

- Monitor the FTP reference value while filling the fuel tank. Refer to **INTRODUCTION - GASOLINE ENGINES** article.
- **Is FTP within specification?**

  YES : Go to A5.

  NO : Go to A4.

A4 MONITOR THE FTP WHILE FILLING THE FUEL TANK WITH THE EVAP SYSTEM DISCONNECTED

- Disconnect the fuel tank-to-EVAP canister quick connect coupling at the EVAP canister.
- Monitor the FTP reference value while filling the fuel tank. Refer to **INTRODUCTION -**

GASOLINE ENGINES article.

- **Is FTP within specification?**

  YES : INSPECT the EVAP system for blockage or restrictions. REPAIR the blockage or restriction. If the blockage or restriction cannot be repaired, INSTALL new EVAP system components.

  NO : Go to A5.

A5 CHECK THE FUEL TANK FILLER PIPE ASSEMBLY FOR BLOCKAGE OR RESTRICTION

- Remove the fuel tank filler pipe assembly. Refer to **FUEL TANK AND LINES** article.
- Inspect the fuel tank filler pipe and fuel tank filler pipe vent tube (if equipped) for a blockage or restriction.
- **Is the fuel tank filler pipe or fuel tank filler pipe vent tube (if equipped) blocked or restricted?**

  YES : If possible, REPAIR the blockage or restriction. If the blockage or restriction cannot be repaired, INSTALL a new fuel tank filler pipe or fuel tank filler pipe vent tube.

  NO : INSTALL a new fuel tank. REFER to **FUEL TANK AND LINES** article.

GENERAL PROCEDURES

FUEL SYSTEM PRESSURE RELEASE

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1. Remove the LH rear quarter trim panel. For additional information, refer to **INTERIOR TRIM AND ORNAMENTATION** article.

NOTE: **The inertia fuel shutoff (IFS) switch is located behind a shield and requires the use of a small screwdriver to release the electrical connector.**

2. Disconnect the IFS switch electrical connector.

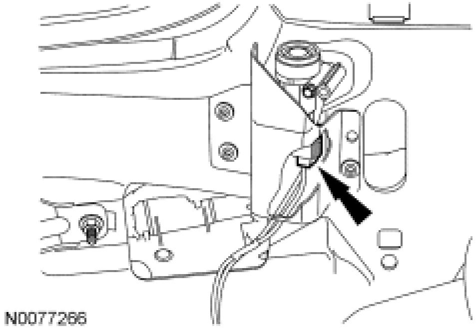


Fig. 1: Identifying IFS Switch Electrical Connector
Courtesy of FORD MOTOR CO.

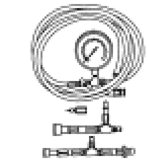
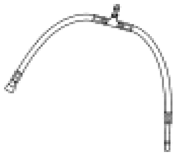
3. Start the engine and allow to idle until the engine stalls.
4. After the engine stalls, crank the engine for approximately 5 seconds to make sure the fuel injection supply manifold pressure has been released.
5. Turn the ignition switch to the OFF position.
6. When the fuel system service is complete, connect the IFS switch electrical connector.

NOTE: **It may take more than one key cycle to pressurize the fuel system. Cycle the ignition key and wait 3 seconds to pressurize the fuel system. Check for leaks prior to starting the engine.**

7. Start the vehicle and check the fuel system for leaks.

FUEL SYSTEM PRESSURE TEST

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
	Fuel Pressure Test Kit	310-D009 (D95L-7211A)
	Adapter, Fuel Pressure Test	310-180

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when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.

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WARNING: **When working on or near the evaporative emission (EVAP) system, disconnect the battery ground cable from the battery. The EVAP system contains fuel vapor and condensed fuel vapor, so an electrical spark may cause a fire or explosion. Failure to follow this instruction may result in serious personal injury.**

1. Release the fuel system pressure. For additional information, refer to **FUEL SYSTEM PRESSURE RELEASE**.
2. Disconnect the battery ground cable. For additional information, refer to **BATTERY, MOUNTING AND CABLES** article.
3. Remove the air cleaner outlet pipe. For additional information, refer to **INTAKE AIR DISTRIBUTION AND FILTERING** article.
4. Disconnect the fuel tube-to-fuel rail quick connect coupling. For additional information, refer to **QUICK CONNECT COUPLING**.
5. Install the Fuel Pressure Test Adapter and the Fuel Pressure Test Kit between the fuel tube and the fuel rail.

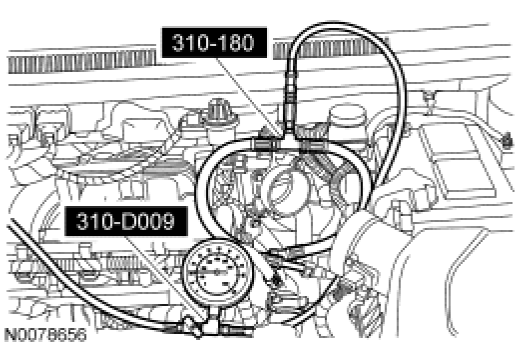


Fig. 2: Identifying Fuel Pressure Test Adapter Fuel Pressure Test Kit
Courtesy of FORD MOTOR CO.

NOTE: **The air cleaner outlet pipe must be installed prior to completing the fuel system pressure test.**

6. Install the air cleaner outlet pipe. For additional information, refer to **INTAKE AIR DISTRIBUTION AND FILTERING** article.

NOTE: **The inertia fuel shutoff (IFS) switch electrical connector was previously disconnected to release the fuel system pressure and must be reconnected to test the fuel system pressure.**

7. Connect the IFS switch electrical connector.
8. Connect the battery ground cable. For additional information, refer to **BATTERY, MOUNTING AND CABLES** article.

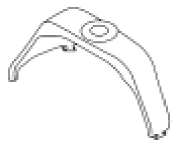
NOTE: **Carry out a key ON engine OFF (KOEO) visual inspection for fuel leaks prior to the Fuel System Pressure Test.**

NOTE: **After completing the fuel system pressure test, open the drain valve on the special tool and release any residual fuel into a suitable container prior to removing the tool.**

9. Test the fuel system pressure to make sure it is within the specified range. For additional information, refer to **SPECIFICATIONS**.

FUEL TANK DRAINING

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
	Fuel Storage Tanker	164-R3202 or equivalent

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All vehicles

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Release the fuel system pressure. For additional information, refer to **FUEL SYSTEM PRESSURE RELEASE**.
3. Disconnect the battery ground cable. For additional information, refer to **BATTERY, MOUNTING AND CABLES** article.
4. Release the fuel tank filler cap and position it aside.
5. Insert a suitable fuel drain tube into the fuel tank filler pipe until it enters the fuel tank.

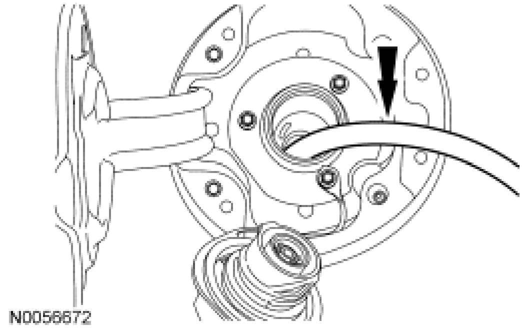

Fig. 3: Inserting Suitable Fuel Drain Tube Into Fuel Tank Filler Pipe
Courtesy of FORD MOTOR CO.

6. Attach the fuel storage tanker to the fuel drain tube and drain as much fuel as possible from the fuel tank.

All wheel drive (AWD) vehicles

7. Remove the rear seats. For additional information, refer to **SEATING** article.
8. Remove the 4 bolts and the fuel level sensor access cover.

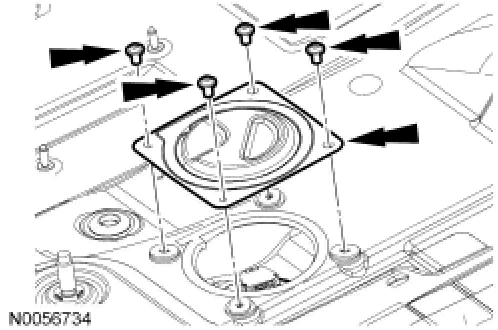

Fig. 4: Identifying Fuel Level Sensor Access Cover & Bolts
Courtesy of FORD MOTOR CO.

NOTE: **Clean the fuel level sensor connection, flange surface and the immediate surrounding area of any dirt or foreign material.**

9. Disconnect the fuel level sensor electrical connector.

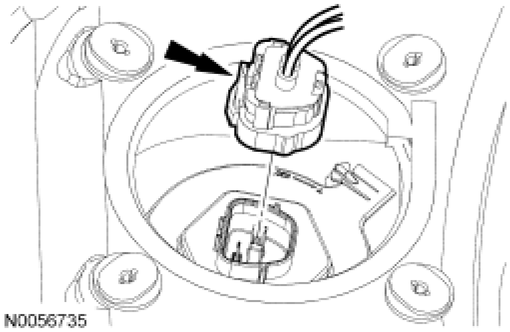

Fig. 5: Identifying Fuel Level Sensor Electrical Connector
Courtesy of FORD MOTOR CO.

NOTE: **Place absorbent pads in the general work area in case of fuel spillage.**

10. Release the lock tab, and using a suitable tool, rotate the fuel level sensor counterclockwise approximately 1/4 turn, lift and position aside.

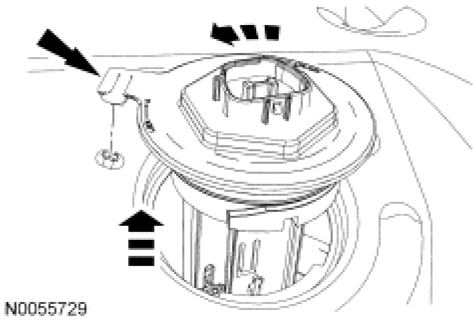

Fig. 6: Rotating Fuel Level Sensor Counterclockwise Approximately 1/4 Turn & Lift Upward
Courtesy of FORD MOTOR CO.

11. Using the special tool and a suitable fuel drain tube, drain the remaining fuel through the fuel level sensor aperture.

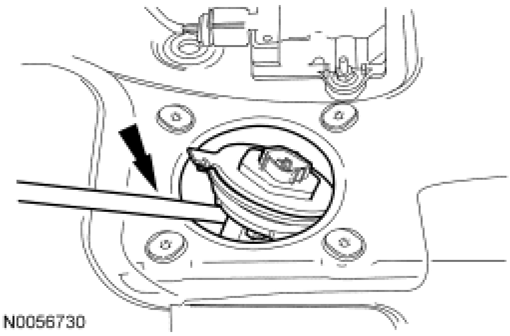

Fig. 7: Draining Remaining Fuel Through Fuel Level Sensor Aperture
Courtesy of FORD MOTOR CO.

QUICK CONNECT COUPLING

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
	Disconnect Tool, Fuel Pipe (5/16")	310-040

MATERIAL

Item	Specification
Motorcraft SAE 5W-20 Premium Synthetic Blend Motor Oil XO-5W20-QSP (US); Motorcraft SAE 5W-20 Super Premium Motor Oil CXO-5W20-LSP12 (Canada); or equivalent	WSS-M2C930-A

Disconnect - Type I

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WARNING: **Before working on or disconnecting any of the fuel tubes or fuel system**

components, relieve the fuel system pressure to prevent accidental spraying of fuel. Fuel in the fuel system remains under high pressure, even when the engine is not running. Failure to follow this instruction may result in serious personal injury.

WARNING: **When working on or near the evaporative emission (EVAP) system, disconnect the battery ground cable from the battery. The EVAP system contains fuel vapor and condensed fuel vapor, so an electrical spark may cause a fire or explosion. Failure to follow this instruction may result in serious personal injury.**

CAUTION: When reusing liquid or vapor tube connectors, make sure to use compressed air to remove any foreign material from the connector retaining clip area before separating from the tube or damage to the tube or connector retaining clip can occur. Apply clean engine oil to the end of the tube before inserting the tube into the connector.

CAUTION: Fuel injection equipment is manufactured to very precise tolerances and fine clearances. It is essential that absolute cleanliness is observed when working with these components or component damage can occur. Always install plugs to any open orifices or tubes.

CAUTION: Do not use any tools. The use of tools may cause a deformity in the clip components which may cause fuel leaks.

1. If servicing a liquid fuel tube quick connect coupling, release the fuel system pressure. For additional information, refer to **FUEL SYSTEM PRESSURE RELEASE**.
2. Disconnect the battery ground cable. For additional information, refer to **BATTERY, MOUNTING AND CABLES** article.
3. Depress the locking tab and release the quick connect coupling from the tube.

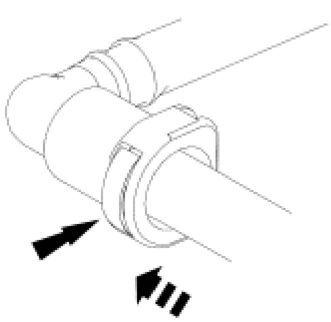

Fig. 8: Locating Fuel Tube Quick Connect Coupling Button
Courtesy of FORD MOTOR CO.

Connect - Type I

NOTE: **Make sure the fuel tube clicks into place when installing the quick connect coupling onto the tube.**

1. Install the quick connect coupling onto the tube until it is fully seated.

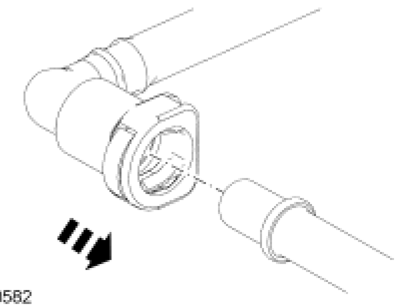

Fig. 9: Installing Quick Connect Coupling Onto Tube
Courtesy of FORD MOTOR CO.

2. Pull on the quick connect coupling and the tube to make sure it is securely fastened.
3. Connect the battery ground cable. For additional information, refer to **BATTERY, MOUNTING AND CABLES** article.

Disconnect - Type II

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1. If servicing a liquid fuel tube quick connect coupling, release the fuel system pressure. For additional information, refer to **FUEL SYSTEM PRESSURE RELEASE**.
2. Disconnect the battery ground cable. For additional information, refer to **BATTERY, MOUNTING AND CABLES** article.
3. Release the quick connect coupling primary locking tab.

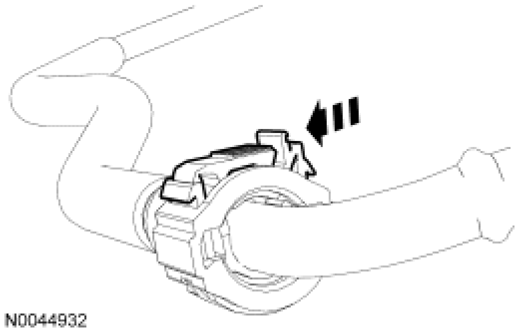

Fig. 10: Releasing Quick Connect Coupling Lock Tab
Courtesy of FORD MOTOR CO.

4. Rotate the primary locking tab to the fully opened position and squeeze the secondary locking tabs to release the locking mechanism.

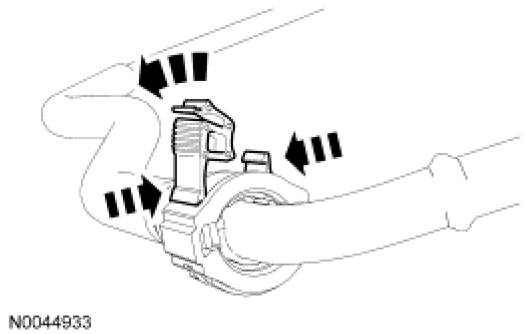

Fig. 11: Releasing Locking Mechanism
Courtesy of FORD MOTOR CO.

5. Push the locking mechanism outward and release the tube.

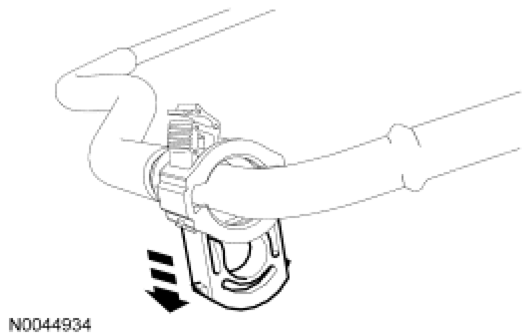

Fig. 12: Pushing Locking Mechanism Outward And Releasing Fuel Supply Tube
Courtesy of FORD MOTOR CO.

6. Remove the quick connect coupling from the tube.

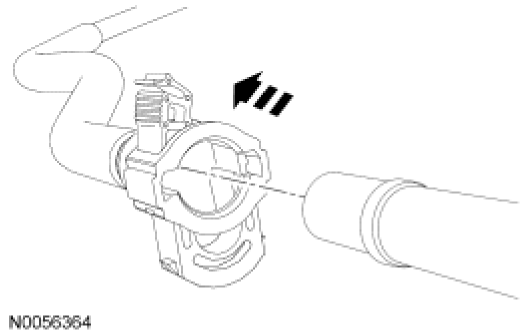

Fig. 13: Removing Quick Connect Coupling From Tube
Courtesy of FORD MOTOR CO.

Connect - Type II

1. Install the quick connect coupling onto the tube until it is fully seated.

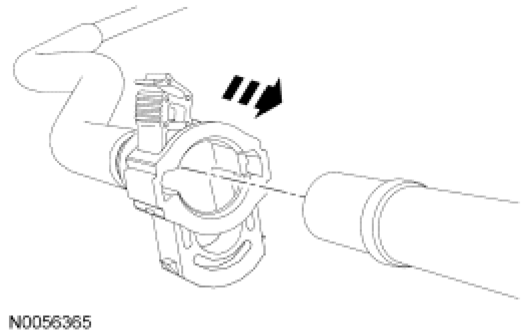

Fig. 14: Installing Quick Connect Coupling Onto Tube
Courtesy of FORD MOTOR CO.

NOTE:	**Make sure the fuel tube clicks into place when installing the quick connect coupling onto the tube.**

2. Depress the locking mechanism until it is flush with the quick connect coupling housing and is fully seated onto the tube.

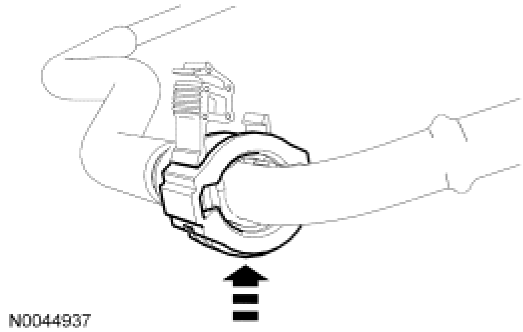

Fig. 15: Depressing Retainer Clip
Courtesy of FORD MOTOR CO.

3. Rotate the primary locking tab on the retainer clip to the closed position.

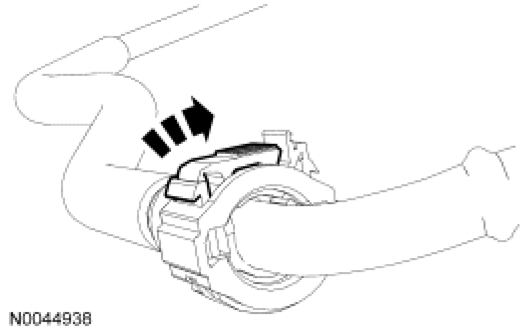

Fig. 16: Rotating Primary Locking Tab On Retainer Clip To Closed Position
Courtesy of FORD MOTOR CO.

4. Pull on the quick connect coupling and the tube to make sure it is securely fastened.
5. Connect the battery ground cable. For additional information, refer to **BATTERY, MOUNTING AND CABLES** article.

Disconnect - Type III

WARNING: **Do not smoke, carry lighted tobacco or have an open flame of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.**

WARNING: **Do not carry personal electronic devices such as cell phones, pagers or audio equipment of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.**

WARNING: **Before working on or disconnecting any of the fuel tubes or fuel system components, relieve the fuel system pressure to prevent accidental spraying of fuel. Fuel in the fuel system remains under high pressure, even when the engine is not running. Failure to follow this instruction may result in serious personal injury.**

WARNING: **When working on or near the evaporative emission (EVAP) system, disconnect the battery ground cable from the battery. The EVAP system contains fuel vapor and condensed fuel vapor, so an electrical spark may cause a fire or explosion. Failure to follow this instruction may result in serious personal injury.**

CAUTION: When reusing liquid or vapor tube connectors, make sure to use compressed air to remove any foreign material from the connector retaining clip area before separating from the tube or damage to the tube or connector retaining clip can occur. Apply clean engine oil to the end of the tube before inserting the tube into the connector.

CAUTION: Fuel injection equipment is manufactured to very precise tolerances and fine clearances. It is essential that absolute cleanliness is observed when working with these components or component damage can occur. Always install plugs to any open orifices or tubes.

CAUTION: Do not use any tools. The use of tools may cause a deformity in the clip components which may cause fuel leaks.

1. If servicing a liquid fuel tube quick connect coupling, release the fuel system pressure. For additional

information, refer to **FUEL SYSTEM PRESSURE RELEASE**.

2. Disconnect the battery ground cable. For additional information, refer to **BATTERY, MOUNTING AND CABLES** article.
3. Squeeze the quick connect coupling retainer locking tabs to release the locking mechanism.

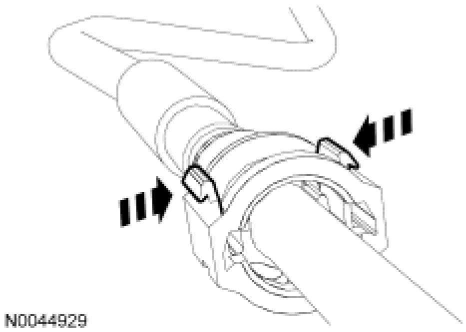

Fig. 17: Releasing Locking Mechanism
Courtesy of FORD MOTOR CO.

4. Push the locking mechanism outward to release the tube.

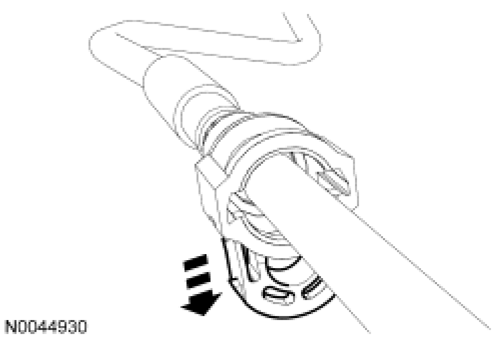

Fig. 18: Pushing Locking Mechanism Outward To Releasing Tube
Courtesy of FORD MOTOR CO.

5. Remove the quick connect coupling from the tube.

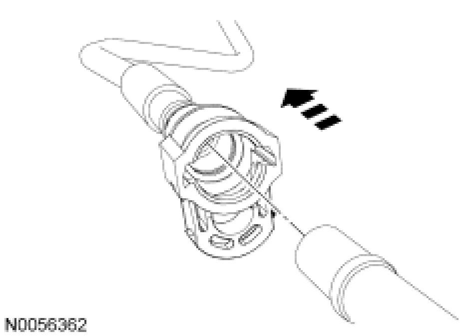

Fig. 19: Removing Quick Connect Coupling From Tube
Courtesy of FORD MOTOR CO.

Connect - Type III

NOTE: **Make sure the fuel tube clicks into place when installing the quick connect coupling onto the tube.**

1. Install the quick connect coupling onto the tube.

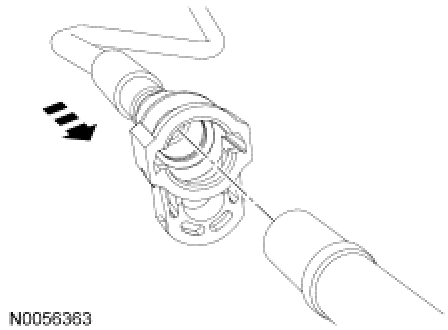

Fig. 20: Installing Quick Connect Coupling Onto Tube
Courtesy of FORD MOTOR CO.

2. Depress the quick connect coupling locking mechanism into the locked position.

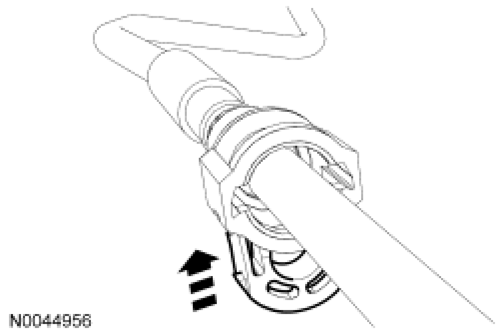

Fig. 21: Depressing Quick Connect Coupling Locking Mechanism Into Locked Position
Courtesy of FORD MOTOR CO.

3. Visually inspect and verify that the locking mechanism is flush with the quick connect coupling housing and that the locking tabs are securely fastened.

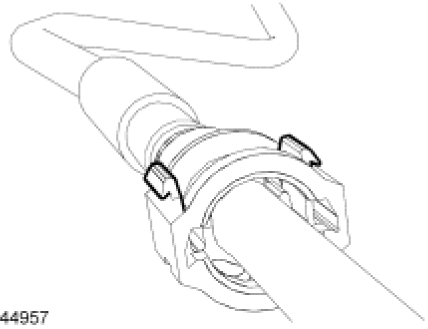

Fig. 22: Identifying Locking Mechanism And Quick Connect Coupling Housing
Courtesy of FORD MOTOR CO.

4. Pull on the quick connect coupling and the tube to make sure it is securely fastened.
5. Connect the battery ground cable. For additional information, refer to **BATTERY, MOUNTING AND CABLES** article.

Disconnect - Type IV

WARNING: **Do not smoke, carry lighted tobacco or have an open flame of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.**

WARNING: **Do not carry personal electronic devices such as cell phones, pagers or audio equipment of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.**

WARNING: **Before working on or disconnecting any of the fuel tubes or fuel system components, relieve the fuel system pressure to prevent accidental spraying of fuel. Fuel in the fuel system remains under high pressure, even when the engine is not running. Failure to follow this instruction may result in serious personal injury.**

WARNING: **When working on or near the evaporative emission (EVAP) system, disconnect the battery ground cable from the battery. The EVAP system contains fuel vapor and condensed fuel vapor, so an electrical spark may cause a fire or explosion. Failure to follow this instruction may result in serious personal injury.**

CAUTION: When reusing liquid or vapor tube connectors, make sure to use compressed air to remove any foreign material from the connector retaining clip area before separating from the tube or damage to the tube

or connector retaining clip can occur. Apply clean engine oil to the end of the tube before inserting the tube into the connector.

CAUTION: Fuel injection equipment is manufactured to very precise tolerances and fine clearances. It is essential that absolute cleanliness is observed when working with these components or component damage can occur. Always install plugs to any open orifices or tubes.

1. If servicing a liquid fuel tube quick connect coupling, release the fuel system pressure. For additional information, refer to **FUEL SYSTEM PRESSURE RELEASE**.
2. Disconnect the battery ground cable. For additional information, refer to **BATTERY, MOUNTING AND CABLES** article.
3. Install the special tool on the tube and push into the quick connect coupling locking clip to release the locking tabs.

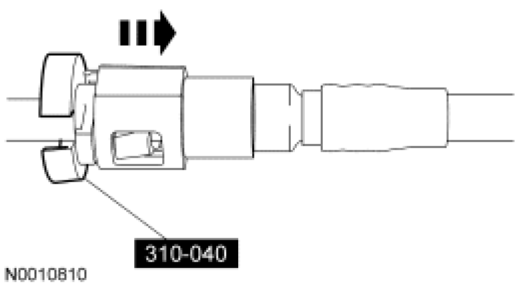

Fig. 23: Disconnecting Quick Connect Coupling From Tube Using Special Tool (310-040)
Courtesy of FORD MOTOR CO.

4. Remove the quick connect coupling from the tube.

Connect - Type IV

NOTE: Make sure the fuel tube clicks into place when installing the quick connect coupling onto the tube.

1. Install the quick connect coupling onto the tube until it is fully seated.

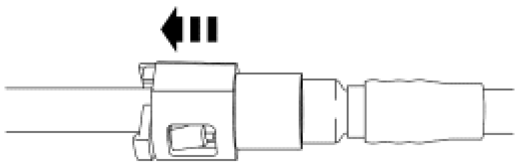

Fig. 24: Installing Quick Connect Coupling Onto Tube Until Fully Seated
Courtesy of FORD MOTOR CO.

2. Pull on the quick connect coupling and the tube to make sure it is securely fastened.
3. Connect the battery ground cable. For additional information, refer to **BATTERY, MOUNTING AND CABLES** article.

Disconnect - Type V

WARNING: **Do not smoke, carry lighted tobacco or have an open flame of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.**

WARNING: **Do not carry personal electronic devices such as cell phones, pagers or audio equipment of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.**

WARNING: **Before working on or disconnecting any of the fuel tubes or fuel system components, relieve the fuel system pressure to prevent accidental spraying of fuel. Fuel in the fuel system remains under high pressure, even when the engine is not running. Failure to follow this instruction may result in serious personal injury.**

WARNING: **When working on or near the evaporative emission (EVAP) system, disconnect the battery ground cable from the battery. The EVAP system contains fuel vapor and condensed fuel vapor, so an electrical spark may cause a fire or explosion. Failure to follow this instruction may result in serious personal injury.**

CAUTION: When reusing liquid or vapor tube connectors, make sure to use compressed air to remove any foreign material from the connector retaining clip area before separating from the tube or damage to the tube

or connector retaining clip can occur. Apply clean engine oil to the end of the tube before inserting the tube into the connector.

CAUTION: Fuel injection equipment is manufactured to very precise tolerances and fine clearances. It is essential that absolute cleanliness is observed when working with these components or component damage can occur. Always install plugs to any open orifices or tubes.

CAUTION: Do not use any tools. The use of tools may cause a deformity in the clip components which may cause fuel leaks.

1. Disconnect the battery ground cable. For additional information, refer to **BATTERY, MOUNTING AND CABLES** article.
2. Squeeze the outer edges of the quick connect coupling to release the locking tabs.

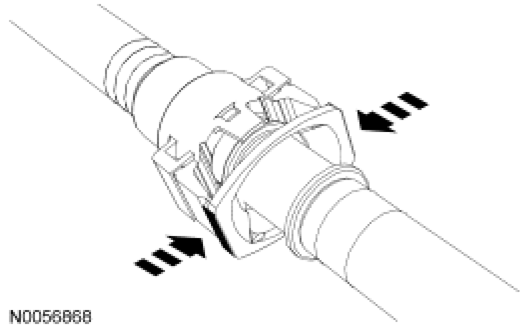

Fig. 25: Squeezing Outer Edges Of Quick Connect Coupling To Release Locking Tabs
Courtesy of FORD MOTOR CO.

3. Remove the quick connect coupling from the tube.

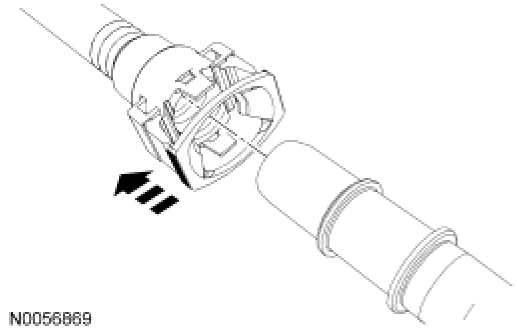

Fig. 26: Removing Quick Connect Coupling From Tube
Courtesy of FORD MOTOR CO.

Connect - Type V

NOTE: **Make sure the fuel tube clicks into place when installing the quick connect coupling onto the tube.**

1. Install the quick connect coupling onto the tube until it is fully seated.

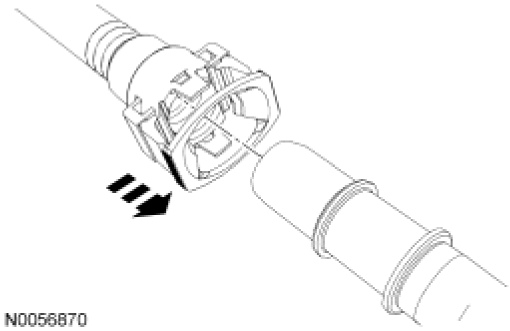

Fig. 27: Installing Quick Connect Coupling Onto Tube Until It Is Fully Seated
Courtesy of FORD MOTOR CO.

2. Pull on the quick connect coupling and the tube to make sure it is securely fastened.
3. Connect the battery ground cable. For additional information, refer to **BATTERY, MOUNTING AND CABLES** article.

Disconnect - Type VI

WARNING: **Do not smoke, carry lighted tobacco or have an open flame of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.**

WARNING: **Do not carry personal electronic devices such as cell phones, pagers or audio equipment of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.**

WARNING: **Before working on or disconnecting any of the fuel tubes or fuel system components, relieve the fuel system pressure to prevent accidental spraying of fuel. Fuel in the fuel system remains under high pressure, even when the engine is not running. Failure to follow this instruction may result in serious personal injury.**

WARNING: **When working on or near the evaporative emission (EVAP) system, disconnect the battery ground cable from the battery. The EVAP system contains fuel vapor and condensed fuel vapor, so an electrical spark may cause a fire or explosion. Failure to follow this instruction may result in**

serious personal injury.

CAUTION: When reusing liquid or vapor tube connectors, make sure to use compressed air to remove any foreign material from the connector retaining clip area before separating from the tube or damage to the tube or connector retaining clip can occur. Apply clean engine oil to the end of the tube before inserting the tube into the connector.

CAUTION: Fuel injection equipment is manufactured to very precise tolerances and fine clearances. It is essential that absolute cleanliness is observed when working with these components or component damage can occur. Always install plugs to any open orifices or tubes.

CAUTION: Do not use any tools. The use of tools may cause a deformity in the clip components which may cause fuel leaks.

1. Release the lock tab on the quick connect coupling.

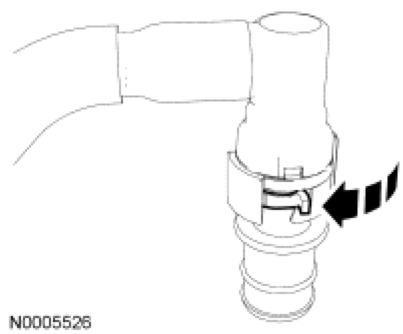

Fig. 28: Positioning Locking Tab Into Unlatched Position
Courtesy of FORD MOTOR CO.

2. Separate the quick connect coupling from the fitting.

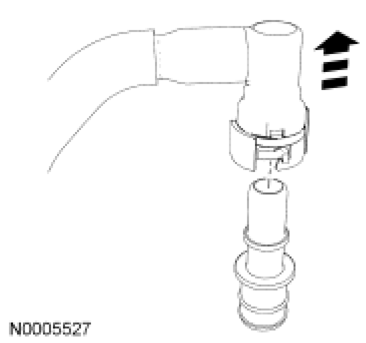

Fig. 29: Identifying Quick Connect Coupling
Courtesy of FORD MOTOR CO.

Connect - Type VI

1. Release the lock tab and install the quick connect coupling onto the fitting.

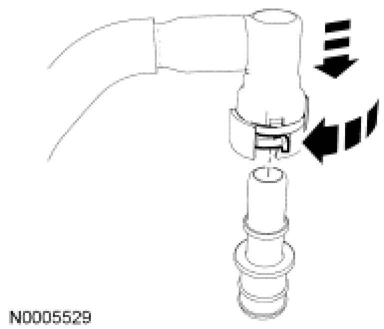

Fig. 30: Identifying Quick Connect Coupling
Courtesy of FORD MOTOR CO.

2. Apply the lock tab into the latched position.

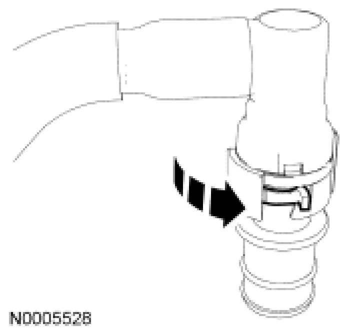

Fig. 31: Positioning Locking Tab Into Latched Position
Courtesy of FORD MOTOR CO.

3. Pull on the quick connect coupling to make sure it is securely fastened.